Empirical Software Engineering manuscript No.
(will be inserted by the editor)

Replicating Posterior Guarantees of Configuration Sampling Strategies under Measured Acquisition Cost

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Received: date / Accepted: date

Abstract Practitioners of configurable software systems rely on budgeted sampling strategies to estimate worst-case (or best-case) performance bounds. A recent FSE 2026 study introduced *posterior risk* and *posterior confidence* as learner-free, scenario-theoretic guarantees for such sampled bounds, and found that Statistical Recursive Search (SRS) consistently yields tighter guarantees than alternative strategies at the same sample size. We replicate this study on its original artifact and extend it to five further dataset families—851 decision scopes across six corpora—and then re-index every decision by *measured acquisition cost* instead of sample size. The guarantee framework replicates everywhere: posterior-risk distributions discriminate strategies in about 95% of scopes, robust to false-discovery-rate correction. The strategy recommendation, however, is population-relative: SRS dominates on the original systems and on an x264 corpus, but is never the best cost-normalized choice on Linux kernel or hyperparameter-optimization pools. Under measured cost, the guarantee/cost Pareto frontier differs from the sample-size frontier in 97.5% of scopes. Two equal-spend experiments explain these boundaries: cost-aware sampling helps exactly when acquisition cost is decoupled from the bounded target (its verdict flips from $6.6\times$ worse to better than random as the coupling weakens), and SRS's advantage requires budget to amortize. Posterior guarantees are thus a sound and portable evaluation framework, but strategy recommendations—including which strategy is "cost-effective"—depend on the outcome, the budget scale, and the cost–target coupling, none of which sample size can express.

Keywords Configurable systems · Sampling strategies · Performance bounds · Scenario theory · Replication · Measurement cost

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1 Introduction

Modern software systems expose hundreds to thousands of configuration options, and the chosen configuration can change performance—execution time, energy consumption, binary size—by orders of magnitude (Xu et al., 2015; Apel et al., 2013; Pereira et al., 2021). For many engineering tasks the question is not what performance a *typical* configuration exhibits, but how bad (or how good) performance can get: worst-case execution time for safety arguments, best-case throughput for capacity planning, largest kernel size for resource-constrained deployment. Exhaustive measurement is infeasible even for moderately sized spaces, so practitioners sample: they measure a budgeted subset of configurations and use the observed extremum as a proxy for the true bound.

How trustworthy is such a sampled bound? A recent FSE 2026 study by Weis et al. (2026b) gives this question a rigorous footing. Building on scenario theory (Calafiore and Campi, 2006; Campi and Garatti, 2008), it defines the *posterior risk* of a sampled bound—the empirical probability that an unseen configuration violates it—and the *posterior confidence* that the risk stays below a tolerance. These measures are computed directly from measurements, with no learned model anywhere in the loop, and they turn strategy comparison into a quantitative exercise: at the same sample size, which strategy delivers lower risk and higher confidence? Across five strategies and seven real-world configurable systems, Weis et al. (2026b) find that Statistical Recursive Search (SRS) (Oh et al., 2017) "yields consistently tighter best-case guarantees—higher posterior confidence and lower posterior risk—than the alternatives at the same budget," while plain solver-based sampling proves least reliable.

This paper starts from a construct-validity observation about the phrase *at the same budget*. Like virtually all sampling studies of configurable systems (Kaltenecker et al., 2019; Pereira et al., 2020; Guo et al., 2018), the FSE study equates budget with *sample size*: strategies are compared after acquiring the same number of configurations. But configurations are not interchangeable units of effort. Measuring a slow, energy-hungry, or large configuration costs more—in wall-clock time, in joules, in compute dollars—than measuring a cheap one, and sampling strategies differ systematically in which configurations they select; indeed, an entire line of strategy design exploits exactly this asymmetry (Sarkar et al., 2015; Nair et al., 2017). Whether strategy *evaluations*, and the recommendations built on them, survive the switch from counted samples to measured spend has not been examined for guarantee-oriented sampling—and guarantees are arguably where the question matters most, because a practitioner who buys a worst-case bound is typically operating under an explicit measurement budget.

Posterior guarantees are also the sharpest possible place to test such a cost argument. Studies whose outcome is prediction error entangle the sampling strategy with a learner, and conclusions can shift with the choice of regressor (Kaltenecker et al., 2020). Posterior risk has no learner: sample, take the extremum, count violations. Any decision change we observe when re-indexing by cost is attributable to the cost construct alone (Figure 1 illustrates the argument).

We therefore conduct a large-scale conceptual replication and extension of Weis et al. (2026b), organized around two research questions:

RQ$_1$: *Do the FSE 2026 core claims replicate beyond the original artifact, at the paper's own fixed-sample-size lens?* We decompose this into four sub-questions mirroring the original claims: discriminative power of posterior risk, monotone improvement with budget, SRS superiority, and solver-based unreliability.

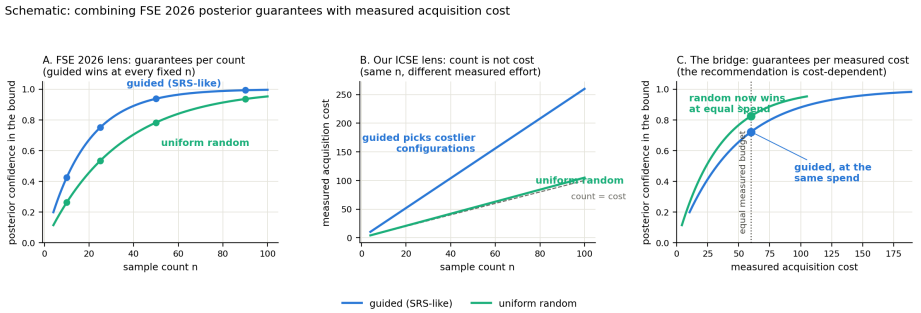


Fig. 1 Schematic of the studied question. (A) At a fixed sample count, a guided strategy such as SRS achieves better guarantees than uniform random sampling—the FSE 2026 lens. (B) The same sample count can correspond to very different measured acquisition costs. (C) Re-indexing guarantees by measured cost changes the decision surface: at an equal measured budget, the recommendation can flip.

RQ$_2$: *Do the FSE 2026 metrics survive re-indexing by measured sampling cost instead of sample size?* We decompose this along a cost-validity lens: cost heterogeneity at fixed size, calibration of size as a cost proxy, impact on recommendations and Pareto frontiers, cost-normalized strategy choice, and an explicit equal-spend protocol in which strategies literally spend the same measured budget.

Our study spans six corpora: the original FSE artifact (direct data reuse, seven systems), the x264 corpus of the ICPE'20 replication study (Pereira et al., 2020), fourteen systems with measured energy and runtime (Weber et al., 2023), nine systems across all ninety-three workloads of the ICSE'23 workload study (Mühlbauer et al., 2023), seven Linux kernel releases (Martin et al., 2022), and thirty-five hyperparameter-optimization pools (Zimmer et al., 2021)—851 (context $\times$ budget) decision scopes in total.

Findings. The guarantee *framework* replicates everywhere: posterior-risk distributions discriminate strategies in $\approx$95% of scopes, robust to false-discovery-rate correction. The strategy *recommendation* replicates only within the original population: SRS dominates on the original systems and on x264, but is never the best cost-normalized choice on the Linux and hyperparameter-optimization pools, and rarely so on the energy and workload families. Under measured cost, the guarantee/cost Pareto frontier differs from the sample-size frontier in 97.5% of the 851 scopes—the single most robust effect in our data—while the single top recommendation changes in 27%. Two dedicated equal-spend experiments explain the boundaries mechanistically: a cost-aware strategy designed to stretch measurement budgets is counterproductive for bounds exactly when acquisition cost correlates with the target (its equal-spend verdict flips from 6.6$\times$ worse than random to *better* than random as the coupling weakens), and SRS's advantage requires budget to amortize (from 4% of scopes at the smallest budgets to 77% at the largest).

Contributions.

– A conceptual replication of Weis et al. (2026b) at unprecedented scale: six corpora, 851 decision scopes, with exact artifact reuse for the original systems and an audited reimplementation of SRS (behaviorally indistinguishable from the artifact, Mann–Whitney $p = 0.95$).

- A systematic re-indexing of posterior guarantees by measured acquisition cost—a construct-validity question that, to our knowledge, had not been examined for guarantee outcomes—at zero additional measurement cost, since every corpus already ships the required cost data.
- Two equal-spend experiments (250 scopes) that identify the governing mechanisms: cost–target coupling and budget amortization. Together they show that "cost-effective" is a property of the triple (strategy, outcome, cost–target coupling), not of a strategy alone.
- A fully scripted, checksummed replication package: per-corpus result tables, hypothesis tests with false-discovery-rate control, and a command-level manifest.

The remainder of the paper follows the structure of our research questions: background and replicated claims (Section 2), study design (Section 3), replication results (Section 4), cost re-indexing results (Section 5), discussion (Section 6), threats (Section 7), related work (Section 8), and conclusions (Section 9).

2 Background and Replicated Claims

2.1 Sampling configuration spaces

A configurable system with k options spans a configuration space whose valid members $\mathscr{V}$ are constrained by a variability model (Apel et al., 2013). A *sampling strategy* selects a subset $S \subseteq \mathscr{V}$, $|S| = n$, whose measurements serve a downstream purpose: training a performance model (Guo et al., 2013; Siegmund et al., 2015; Ha and Zhang, 2019), finding a near-optimal configuration (Oh et al., 2017; Nair et al., 2020), or, in our setting, bounding worst-case (or best-case) performance. Strategy families include uniform random sampling (in practice often solver-mediated (Plazar et al., 2019; Heradio et al., 2022)), coverage oriented t-wise sampling (Medeiros et al., 2016; Baranov and Legay, 2024; Heß et al., 2024), distance- and diversity-based sampling (Kaltenecker et al., 2019), and adaptive strategies that use intermediate measurements to steer selection (Sarkar et al., 2015; Nair et al., 2017).

Statistical Recursive Search (SRS) (Oh et al., 2017), the strategy recommended by Weis et al. (2026b), is adaptive: it repeatedly draws a small random batch (25 configurations), identifies option decisions shared by the two best configurations of the batch, keeps the decisions whose enabled/disabled performance difference is significant under Welch's t-test ($p < 0.05$), restricts the space accordingly, and recurses; if recursion stops before the budget is exhausted, the remainder is filled with random configurations.

2.2 Posterior risk and posterior confidence

Let $y(c)$ be the measured target of configuration $c \in \mathscr{V}$ and let S be a sample. For a worst-case (maximum) bound, the sampled bound is $\hat{b} = \max_{c \in S} y(c)$, and the *posterior risk* is the fraction of the reference universe that violates it:

$$R(S) \;=\; \frac{\left|\{\, c \in \mathscr{V} : y(c) > \hat{b} \,\}\right|}{|\mathscr{V}|}. \tag{1}$$

Repeating the sampling r times yields an empirical distribution of risks; the *posterior confidence* at tolerance ε is the fraction of repetitions with $R \leq \varepsilon$, i.e., the empirical CDF of risk. Integrating this CDF over $\varepsilon \in [0,1]$ gives the *integrated posterior confidence* $A_{\text{post}} = 1 - \overline{R}$,

a scalar summary (higher is better). Scenario theory provides a strategy-agnostic *prior* baseline: for uniform random samples of size N, confidence at tolerance ε is $1-(1-\varepsilon)^{N+1}$ (we follow the artifact's exponent convention and report a sensitivity with exponent N). The *integrated confidence discrepancy* $\omega = A_{\text{post}} - A_{\text{prior}}$ measures how much better (or worse) a strategy's guarantees are than the random-sampling prior at the same size. All quantities are computed from measurements only; no learner is involved.

Weis et al. (2026b) evaluate five strategies (uniform random, solver-based, diversified distance-based, t-wise, SRS) on seven systems with exhaustively measured configuration spaces (7z, BerkeleyDB-C, Dune, Hipacc, JavaGC, LLVM, Polly), at a per-system budget $t2$ matching the size of a pairwise-coverage sample (56–843 configurations), with 100 repetitions. We refer to their headline results as four claims:

C1 (discriminative power). Posterior risk and confidence distinguish sampling strategies at the same budget, enabling rigorous comparison beyond observed performance values.
C2 (SRS superiority). SRS yields consistently tighter guarantees—lower posterior risk, higher posterior confidence—than the alternatives at the same budget.
C3 (solver unreliability). Plain solver-based sampling is the least reliable strategy for probabilistic guarantees.
C4 (budget scaling). Guarantees improve as the sample size grows.

2.3 Sample size versus sampling cost

The budget in C1–C4 is a *count*. The construct-validity concern is that counts are not effort: measuring configuration c costs some $\kappa(c) > 0$ (seconds of wall-clock, joules, dollars), and strategies select different configurations. We quantify the gap with the *cost share* $\%cost(S) = \sum_{c\in S}\kappa(c)/\sum_{c\in\mathscr{V}}\kappa(c)$, the *size share* $\%n(S) = |S|/|\mathscr{V}|$, and their ratio, the *calibration ratio* $\%cost/\%n$ (1.0 iff the sample is cost-representative). Cost-indexing the prior baseline is then immediate: a strategy that spent cost share $\%cost$ has an *effective cost-equivalent size* $N_{\text{eff}} = \%cost \cdot |\mathscr{V}|$, and $\omega_{\text{cost}} = A_{\text{post}} - A_{\text{prior}}(N_{\text{eff}})$ asks whether its guarantees beat what uniform random sampling would be expected to deliver *for the same spend*. Analogously, Pareto frontiers over (confidence, budget) can use either n or measured cost as the budget axis; comparing the two frontiers (by membership and Jaccard overlap) quantifies how much the choice of cost construct changes the decision surface. Throughout, κ is itself a measured property of the corpus (runtime or energy), not a model.

2.4 Replication terminology

Following Shull et al. (2008) and Gómez et al. (2014), RQ_1 is a *close* replication on the original artifact (same data, same strategy samples, independently reimplemented analysis) and a *conceptual* replication on five further corpora (same constructs and hypotheses, different populations, partially different strategy sets). RQ_2 is an *extension*: it changes one construct—the budget axis—while holding everything else fixed.

Table 1 Corpora of the study. "Scopes" are (context × budget level) decision cells; κ is the measured acquisition-cost property.

Corpus (role)	Contexts	Budgets	Scopes	Cost κ
FSE artifact (close replication)	7 systems	$t2$	7	runtime
x264 / ICPE'20 (replication)	17 inputs × 2 targets	$t1$–$t3$ (fixed $t2$)	34	encoding time
ENERGY / Twins (extension)	14 systems × 4 cost/target pairs	$n8$–$n64$	220	energy or runtime
WORKLOAD / ICSE'23 (extension)	9 systems, all 93 workloads	$n8$–$n64$	359	wall time
Linux / TuxKConfig (external validity)	7 releases	1%, 2%, 5% of pool	21	vmlinux size
LCBench (external validity)	35 OpenML datasets	$n8$–$n64$	210	training runtime

3 Study Design

3.1 Corpora

Table 1 summarizes the six corpora. Each corpus contributes *decision scopes*: one scope is a (measurement context, budget level) cell in which strategies are compared. Scopes are the unit of all counts reported in this paper (851 in total).

FSE artifact. We reuse the original artifact (Weis et al., 2026a) directly: its per-system `measurements.csv` universes (1,024–193,536 configurations) and its shipped per-run sample files for all five strategies, including the authors' own SRS runs (100 repetitions per system at the per-system $t2$ budget of 56–843 configurations). This makes RQ_1 on this corpus a close replication with exact data reuse; only the analysis is independently reimplemented. Notably, these seven exhaustively measured systems and the archive strategy families (random, solver-based, distance-based, t-wise, at t-wise-derived budget levels) originate from the ICSE'19 distance-based sampling study (Kaltenecker et al., 2019), which Weis et al. (2026b) build on; our replication therefore simultaneously revisits the measurement population of that earlier study, under a guarantee outcome instead of a prediction-error one. The acquisition cost of a configuration is its measured performance (the time to execute the benchmark that the study measures), so re-indexing by cost requires no additional data.

x264. The ICPE'20 corpus (Pereira et al., 2020) measures 201,600 executions of x264 across 17 input videos with two target properties (encoding time, output size). It ships the same archive-format strategy families as the FSE artifact plus two more (distance-based and Henard's diversified variant), at three t-wise budget levels; encoding time serves as κ for both targets. Following the FSE protocol we fix $t2$ for headline counts (34 scopes) and use all levels for scaling analyses.

ENERGY. The Twins corpus (Weber et al., 2023) measures energy and runtime for 14 systems (2,304–68,640 configurations after revision pinning). Because it measures two cost-like properties, it lets us cross the target and the cost construct: each system contributes four scenarios (target, κ) $\in$ {energy, runtime}^{2}, asking the FSE question under a different *notion* of cost.

WORKLOAD. The ICSE'23 workload-impact corpus (Mühlbauer et al., 2023) measures 9 systems across 93 workload/input contexts. Each (system, workload) pair is a context; the measured wall time (throughput for the h2 database, with inverse orientation) is both target and κ. This family probes whether guarantee-per-cost decisions are stable across workloads of the same system.

Linux and LCBench (external validity). The TuxKConfig pools (Martin et al., 2022) cover seven kernel releases (25k–92k measured configurations each, ≈10k options); the target and κ are the built kernel size. LCBench (Zimmer et al., 2021) provides 35 hyperparameter-

optimization pools of 2,000 configurations each; the target is final validation accuracy and κ is training runtime. Both are *measured pools*, not exhaustive spaces: all posterior-risk statements for them are pool-relative. LCBench additionally leaves the configurable-software population entirely—a deliberate stress test, and the corpus where cost and target are most weakly coupled.

3.2 Strategies

The two archive-based corpora (FSE artifact, x264) use the strategy samples shipped with their artifacts: uniform random, solver-based, diversified distance-based, t-wise (x264 adds distance-based and Henard's variant), plus SRS. The four remaining corpora use a generated-sampling protocol with five deployable strategies: uniform random, balanced (option-frequency balancing), diversity (max–min distance space filling), pairwise (greedy pair coverage), and pilotCost (a cost-aware strategy that spends 20% of the budget on a random pilot, fits a ridge model *of the cost*, and diversifies among predicted-cheap configurations); Linux replaces pairwise/pilotCost with coverage and uncertainty (residual-driven active learning) samplers suited to its scale. SRS is generated natively on all these corpora. Diagnostic oracle strategies (cheapest-/most-expensive-first) are excluded from all headline counts.

3.3 SRS audit

Because SRS carries the headline recommendation of Weis et al. (2026b), we audited our reimplementation against the artifact notebook line by line and behaviorally. All protocol constants match (batch size 25, two-sided Welch $p < 0.05$, worst-case orientation, random fill, budget clamp). On a synthetic system (3,000 configurations, 40 seeds), the posterior-risk distributions of the two implementations are statistically indistinguishable (Mann–Whitney $p = 0.95$), and both dominate uniform random ($p < 10^{-5}$), reproducing the paper's headline in vitro. The audit surfaced one discrepancy worth recording: the artifact's random fill draws from the current recursion pool *including* already-selected configurations, so its samples can contain duplicates; deduplicated, its shipped $t2$ SRS samples are 1–5 configurations short of nominal size in 2–42% of runs per system. Our implementation never duplicates (the conservative choice for cost accounting); the FSE-artifact corpus results use the artifact's own samples regardless, so RQ_1 on that corpus is unaffected by reimplementation choices.

3.4 Protocol decisions

All runs use the worst-case orientation (maximum bound); the prior confidence uses the artifact's exponent convention $N+1$ (a full rerun with exponent N changes no decision in any corpus). Stochastic strategies use 100 seeds on the replication corpora and Linux, 30 on the generated families and LCBench; deterministic strategies (diversity, pairwise) contribute one run per scope. Budget grids are capped at $\min(30\% \text{ of the pool}, 256)$ configurations for the generated families (four levels, n=8–64) and at 1%/2%/5% of the pool for Linux. All pipelines are seeded and deterministic end to end; as an incidental check, an interrupted Linux run reproduced its first release bit-identically when resumed three days later.

3.5 Hypotheses, statistics, and decision metrics

Within each scope we test, following the FSE-style protocol: **H1** posterior-risk distributions differ across strategies (Kruskal–Wallis (Kruskal and Wallis, 1952)); **H5** SRS has lower posterior risk than random and **H6** SRS's acquisition cost differs from random's (Mann–Whitney (Mann and Whitney, 1947), with Vargha–Delaney $\hat{A}_{12}$ effect sizes (Vargha and Delaney, 2000)). Three further outcomes are decision rules rather than tests: **H2** the top recommendation (highest integrated-confidence discrepancy) differs between the size-indexed and cost-indexed baselines; **H3** SRS is the best cost-normalized choice (ω_{cost}, Section 2.3); **H4** the (confidence, budget) Pareto frontier differs between the size axis and the measured-cost axis (flagged when frontier membership differs; quantified by Jaccard overlap). For the significance-based hypotheses we control the false discovery rate at 0.05 with Benjamini–Hochberg within each (corpus, hypothesis) family (Benjamini and Hochberg, 1995) and report corrected counts wherever correction changes them.

3.6 Equal-spend protocol

The ω_{cost} handicap compares a strategy's guarantees to the random-sampling prior *at the size its spend would buy*. To verify that conclusions are not an artifact of this counterfactual, we add a literal protocol: fix a budget $B = n_{\text{ref}} \cdot \bar{\kappa}$ (uniform random's expected spend at reference size $n_{\text{ref}} \in \{8, 16, 32, 64\}$) and let *every* strategy buy as many configurations as B affords (per-seed binary search on its own sample size, capped at the protocol grid; cap hits are censoring and are reported). The outcome is the realized posterior risk at equal measured spend. We run this on ENERGY (110 scopes; strong cost–target coupling) and LCBench (140 scopes; weak coupling), 30 seeds each.

3.7 Reproducibility

Every run is scripted; a command-level manifest records inputs (with checksums), protocol parameters, and outputs. Long-running sweeps are checkpointed per context and resumable. Aggregate result tables, hypothesis tests, and all figures are versioned alongside the code.

4 RQ_1: Does FSE 2026 Replicate at Fixed Sample Size?

This section evaluates the four FSE claims (Section 2.2) at the original paper's own lens: strategies compared after acquiring the same *number* of configurations. Table 2 aggregates the per-corpus outcomes; Figure 2 shows the full outcome matrix, including the cost-indexed outcomes of Section 5.

4.1 RQ1.1: Discriminative power (C1)

On the original artifact, all seven systems show significant differences between the posterior-risk distributions of the five strategies, reproducing C1 exactly. Figure 3 illustrates the replication on Polly, the system the original paper singles out for discussion, in the original paper's own visual idiom: per-strategy posterior-risk distributions over 100 repetitions (left)

Table 2 RQ_1 outcomes per corpus at fixed sample size. H1: KW test on posterior-risk distributions (α=.05; arrows give Benjamini–Hochberg corrected counts where they differ). Best (confidence): SRS has the highest integrated posterior confidence. Best (risk): SRS has the lowest median posterior risk. H5: SRS risk significantly below random.

Corpus	Scopes	H1 differ	SRS best (conf.)	SRS best (risk)	H5 SRS<random
FSE artifact	7	7/7	5/7	4/7	6/7
x264	34	34/34	28/34	24/34	34/34
ENERGY	220	212/220	12/220	22/220	52 → 42/220
WORKLOAD	359	341/359	59/359	72/359	84 → 70/359
Linux	21	21/21	0/21	0/21	5 → 1/21
LCBench	210	154 → 144/210	1/210	0/210	0/210
Total	851	≈95%	—	—	—

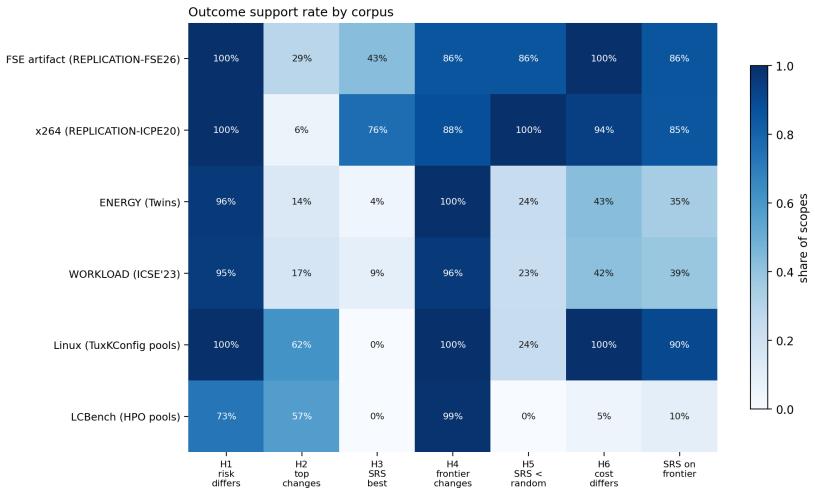


Fig. 2 Outcome support rates by corpus (share of scopes; Linux row shows the full-protocol run). H1/H5/H6 are significance-based; H2–H4 and frontier membership are decision rules (Section 3.5).

and the confidence discrepancy against the scenario-theoretic prior as the accepted risk ε varies (right). The strategies separate cleanly—t-wise trades a deterministic, slightly higher risk for the largest confidence gain at small ε, SRS and solver-based sampling beat the prior, distance-based sampling falls below it. The claim then generalizes to every corpus we add: 34/34 x264 contexts, 212/220 ENERGY scenarios, 341/359 WORKLOAD contexts, 21/21 Linux release-level scopes, and 154/210 LCBench pools (144 after FDR correction; the only corpus where correction moves the count). Overall, posterior risk discriminates strategies in ≈95% of 851 scopes—with no learner in the loop and across targets as different as encoding time, energy, kernel size, and model accuracy.

Finding 1 (C1 replicates and generalizes). Posterior risk and confidence discriminate sampling strategies in ≈95% of 851 scopes across six corpora, robust to FDR correction. The FSE 2026 evaluation framework is sound and portable.

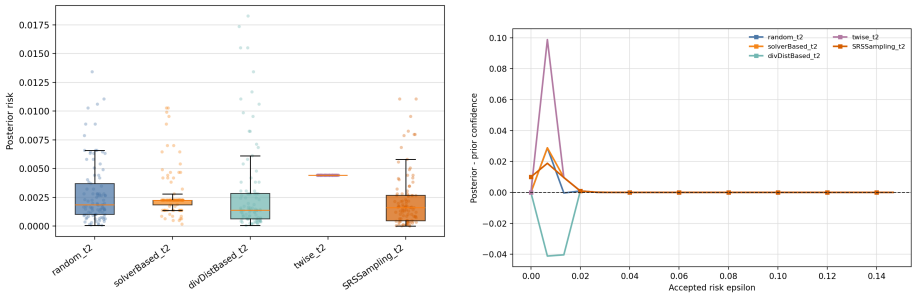


Fig. 3 Close replication on Polly (60,000 configurations, budget $t2 = 345$), in the visual idiom of Weis et al. (2026b). Left: posterior-risk distributions per strategy (100 repetitions). Right: posterior minus prior confidence as a function of the accepted risk ε (positive: better guarantees than the random-sampling prior at the same size).

Table 3 RQ1.2, budget scaling: share of (context, strategy) cells whose guarantees improve monotonically along the budget grid (median risk non-increasing; integrated confidence non-decreasing), for all strategies and for SRS alone. The FSE artifact runs a single budget level and is not applicable.

	All strategies		SRS	
Corpus	risk ↓	conf. ↑	risk ↓	conf. ↑
x264	100%	100%	100%	100%
ENERGY	76%	77%	100%	100%
WORKLOAD	81%	81%	100%	100%
Linux	98%	100%	100%	100%
LCBench	65%	65%	49%	54%

4.2 RQ1.2: Budget scaling (C4)

For each (context, strategy) pair we check whether median posterior risk is non-increasing (and integrated confidence non-decreasing) along the budget grid; Table 3 reports the share of cells where this holds. On configurable-software corpora the FSE expectation holds broadly: 100% of x264 cells, 98% of Linux cells, 81% of WORKLOAD and 76% of ENERGY cells (the generated families use small budgets where single-seed deterministic strategies can be non-monotone). For SRS specifically, scaling is perfect (100%) on *all* configurable-software corpora. On LCBench, however, monotonicity degrades to 65% of cells overall and 49% for SRS: on hyperparameter-optimization pools, growing the sample does not reliably tighten SRS's bound, because its recursive restriction can lock onto hyperparameter regions whose accuracy plateau differs from where the pool's true optimum lies.

Finding 2 (C4 replicates within the population, cracks outside). Guarantees improve monotonically with sample size for SRS on every configurable-software corpus (100% of cells), but only in 49% of LCBench cells—the first boundary condition.

4.3 RQ1.3: SRS superiority (C2)

On the original artifact, SRS is significantly better than uniform random in 6/7 systems (H5), has the highest integrated confidence in 5/7, and the lowest median risk in 4/7—consistent

Table 4 RQ1.3–1.4, fixed-size strategy ranking on the replication corpora: mean rank by median posterior risk across contexts at the $t2$ budget (1 = lowest risk). Strategies absent from a corpus are marked —.

Strategy	FSE artifact (of 5)	x264 (of 7)
SRS	**1.6**	**1.1**
Diversified distance-based	2.3	2.9
Uniform random	3.0	3.3
Distance-based	—	4.3
t-wise	4.1	4.4
Henard's diversified	—	5.1
Solver-based	3.7	6.1

with the original finding that SRS is the strongest, though not uniformly dominant, strategy. On x264 the replication is even stronger: significantly better than random in 34/34 contexts, best integrated confidence in 28/34. x264's budgets are t-wise-scale (dozens to hundreds of samples), matching the original protocol. Table 4 summarizes the fixed-size strategy ranking on both replication corpora: SRS is the top-ranked strategy on both (mean rank 1.6 of 5, and 1.1 of 7), with distance-based sampling a consistent second.

The claim does not travel further. On the generated families at small budgets (n=8–64), SRS is best in only 5–20% of scopes and beats random significantly in ≈19% (FDR-corrected); space-filling strategies (diversity, pairwise) dominate. On Linux, SRS is *never* the best strategy at fixed size (0/21) and beats random in only 1/21 scopes after correction; on LCBench it is best by integrated confidence in a single pool (1/210) and never by risk. Anticipating Section 5: the equal-spend experiment shows this is chiefly a *budget* effect—SRS's advantage over random grows from 4% of ENERGY scopes at n8-scale budgets to 77% at n64-scale (Figure 9B)—which reconciles its dominance at the original protocol's larger budgets with its weakness at ours.

Finding 3 (C2 replicates where FSE tested it; it is budget- and domain-dependent). SRS superiority replicates fully on the original systems and on x264 (34/34), fades at small budgets on the generated families, and vanishes on Linux and LCBench (best in at most 1 of 231 scopes). The advantage requires budget to amortize.

4.4 RQ1.4: Solver unreliability (C3)

Solver-based sampling ranks near the bottom on both corpora where it participates (Table 4): mean median-risk rank 3.7 of 5 strategies on the FSE artifact and clearly worst, 6.1 of 7, on x264. On the FSE artifact only t-wise ranks lower by median risk—but t-wise's deterministic sample yields zero risk variance and hence high confidence at small tolerances, whereas solver-based sampling combines mediocre risk with high variance, matching the original paper's characterization of unreliable guarantees. Its samples, drawn from a SAT solver's biased enumeration order, systematically miss the performance extremes that bounds require, echoing prior findings on solver bias (Plazar et al., 2019).

Finding 4 (C3 replicates). Plain solver-based sampling remains the least reliable strategy for guarantees on both corpora that include it.

4.5 RQ_1 answer

All four FSE claims replicate on the original artifact under exact data reuse, and the framework-level claims (C1, C3) replicate on every applicable corpus. The strategy-level claims (C2, C4) replicate within the population the original study sampled from—configurable software at t-wise-scale budgets—and degrade predictably outside it. No claim reverses on the original systems. In replication terms: a successful close replication, and a conceptual replication that maps the claims' boundary conditions rather than refuting them.

5 RQ_2: Do the Metrics Survive Measured Cost?

We now hold the metrics fixed and change one construct: the budget axis, from sample count to measured acquisition cost (Section 2.3).

5.1 RQ2.1–2.2: Cost heterogeneity and the calibration of size as a proxy

A fixed sample size is not a fixed measured effort. Within-system per-configuration costs vary by up to a coefficient of variation of 1.42 and a $p90/p10$ ratio of 47.9 (BerkeleyDB-C) on the FSE artifact alone. Consequently, the calibration ratio $\%cost/\%n$ of actual samples—1.0 when a sample is cost-representative—is centered near 1.0 in every corpus but with heavy tails: 0.19–3.11 across FSE-artifact cells, 0.03–7.05 on ENERGY, 0.09–2.77 on WORKLOAD, 0.04–2.48 on LCBench, 0.64–2.57 on Linux, and 0.62–1.43 on x264 (Figure 4). In other words, sample size is an approximately *unbiased* but *high-variance* proxy for measurement effort: two strategies that both acquire n configurations can differ in actual spend by a factor of two to thirty, and the direction of the deviation is strategy-systematic (e.g., SRS leans expensive on Linux in 21/21 scopes, H6; `pilotCost` leans cheap by construction).

Finding 5. Sample size is an unbiased but high-variance proxy for acquisition cost: same-size samples differ in measured spend by up to an order of magnitude, and the deviations are strategy-systematic—a construct-validity gap for any budgeted strategy comparison.

5.2 RQ2.3: Decision impact

Does the proxy error matter for decisions? Two decision artifacts can change when the budget axis switches from n to measured cost: the single top recommendation (highest ω), and the Pareto frontier over (integrated confidence, budget)—the set of non-dominated (strategy, budget) choices a practitioner should consider. Table 5 and Figure 5 give the counts.

The frontier result is the most robust effect in our data: in 830 of 851 scopes (97.5%), the set of guarantee/cost-efficient choices under measured cost differs from the set under sample count, and not marginally: the median Jaccard overlap between the two frontiers is 0.14–0.33 depending on the corpus. The top-1 recommendation is more stable and strongly corpus-dependent—from 6% of x264 contexts to 62% on Linux: the Pareto-level effect is the robust one, single-winner flips the secondary one.

Even on the original FSE artifact, re-indexing by the artifact's own runtime measurements changes the frontier in 6/7 systems. Table 6 gives the per-system view: cost heterogeneity (columns 3–4) is a property of the system, and where it is large (7z, BerkeleyDB-C),

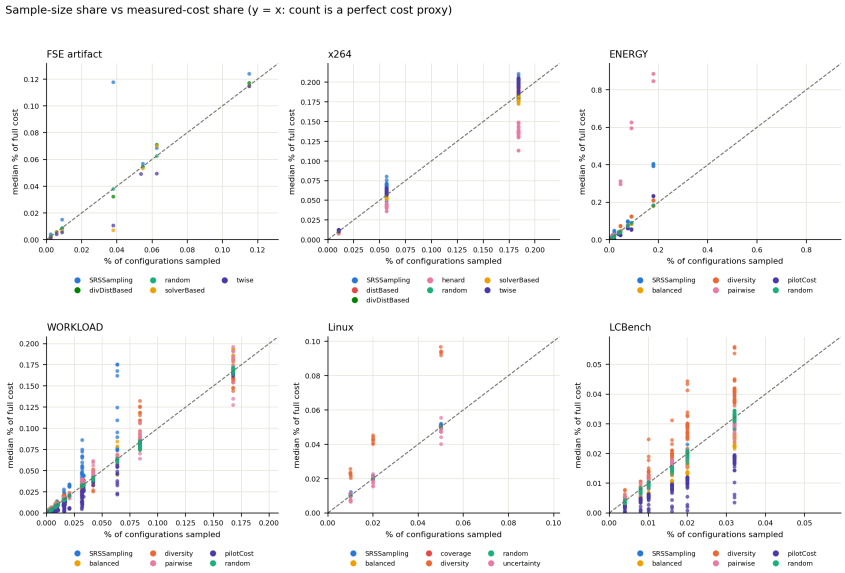


Fig. 4 Share of configurations sampled versus share of total measurement cost, per strategy and budget level (medians across runs). On the $y{=}x$ diagonal, sample size is a perfect cost proxy.

Table 5 RQ_2 decision impact per corpus: scopes where the top recommendation (H2) and the Pareto frontier (H4) differ between the size axis and the measured-cost axis.

Corpus	Scopes	Top changes (H2)	Frontier changes (H4)
FSE artifact	7	2 (29%)	6 (86%)
x264	34	2 (6%)	30 (88%)
ENERGY	220	31 (14%)	220 (100%)
WORKLOAD	359	61 (17%)	346 (96%)
Linux	21	13 (62%)	21 (100%)
LCBench	210	120 (57%)	207 (99%)
Total	851	229 (27%)	**830 (97.5%)**

the size-indexed and cost-indexed winners diverge—SRS wins both axes only on the three systems whose samples happen to be nearly cost-representative. The frontier overlap (last column) is below 0.5 on six of seven systems.

Finding 6. Re-indexing guarantees by measured cost changes the Pareto frontier of defensible (strategy, budget) choices in 97.5% of 851 scopes—including 6/7 original FSE systems—while the single top pick changes in 27%. Sample-size budgets systematically misrepresent the decision surface a cost-aware practitioner faces.

5.3 RQ2.4: Cost-normalized recommendation

Which strategy is best per unit of measured cost, by ω_{cost}? Table 7 counts the cost-normalized winners per corpus. SRS remains the winner where it also won at fixed size—26/34 x264 contexts, 3/7 FSE-artifact systems—but nowhere else: 9/220 ENERGY, 31/359 WORK-

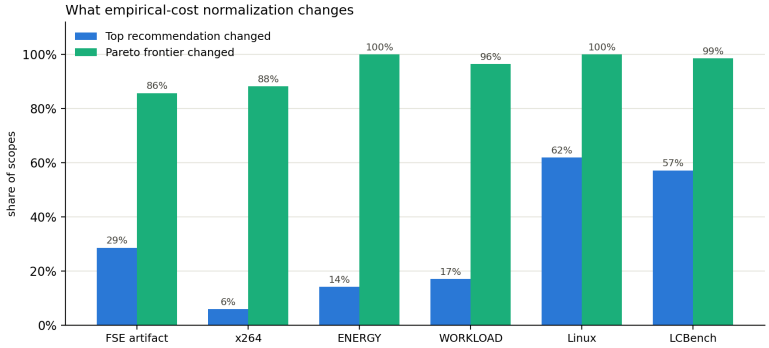


Fig. 5 What switching the budget axis changes, per corpus: the single top recommendation versus the Pareto frontier of defensible choices.

Table 6 Per-system decisions on the original FSE artifact at the $t2$ budget. CV: coefficient of variation of per-configuration cost; $p90/p10$: cost dispersion; best by size/cost: winner by integrated confidence discrepancy under each budget axis; J: Jaccard overlap of the two Pareto frontiers.

System	$\lvert\mathscr{Y}\rvert$	Cost CV	$p90/p10$	Best by size	Best by cost	J
7z	68,640	1.19	10.1	SRS	t-wise	0.50
BerkeleyDB-C	2,560	1.42	47.9	SRS	random	0.20
Dune	2,304	0.34	1.8	SRS	SRS	0.25
Hipacc	13,485	0.44	2.6	distance	distance	0.25
JavaGC	193,536	1.02	7.8	SRS	SRS	0.33
LLVM	1,024	0.07	1.2	distance	distance	1.00
Polly	60,000	0.24	2.1	SRS	SRS	0.20

Table 7 RQ2.4, cost-normalized winners: how often each strategy has the highest integrated-confidence discrepancy against the cost-indexed prior (ω_{cost}), per corpus (winners with at least two scopes shown).

Corpus	Scopes	Cost-normalized winners (scopes won)
FSE artifact	7	SRS (3), distance-based (2), t-wise (1), random (1)
x264	34	SRS (26), Henard's (5), distance-based (3)
ENERGY	220	diversity (108), pairwise (74), balanced (20), SRS (9)
WORKLOAD	359	pairwise (155), diversity (108), balanced (48), SRS (31)
Linux	21	diversity (19), uncertainty (1), balanced (1)
LCBench	210	diversity (69), balanced (62), pilotCost (54), pairwise (25)

LOAD, 0/21 Linux, 0/210 LCBench (Figure 7). On the small-budget families the space-filling strategies dominate the cost-normalized picks (diversity 108 and pairwise 74 of 220 on ENERGY; pairwise 155 and diversity 108 of 359 on WORKLOAD), and on LCBench even the cost-aware pilotCost takes a quarter of the scopes (54/210)—a first hint of the coupling mechanism that Section 5.4 isolates.

Linux crystallizes the distinction between *winning* and *being worth considering*. At full statistical power (100 seeds), SRS is never the best cost-normalized strategy (0/21), its acquisition cost differs significantly from random's in 21/21 scopes—it buys systematically more expensive kernels—and yet it sits *on* the empirical guarantee/cost frontier in 19/21 scopes (Figure 8 illustrates the three regimes: SRS best, SRS on the frontier without winning, SRS dominated). SRS buys real guarantee quality at a real cost premium; the cost-indexed anal-

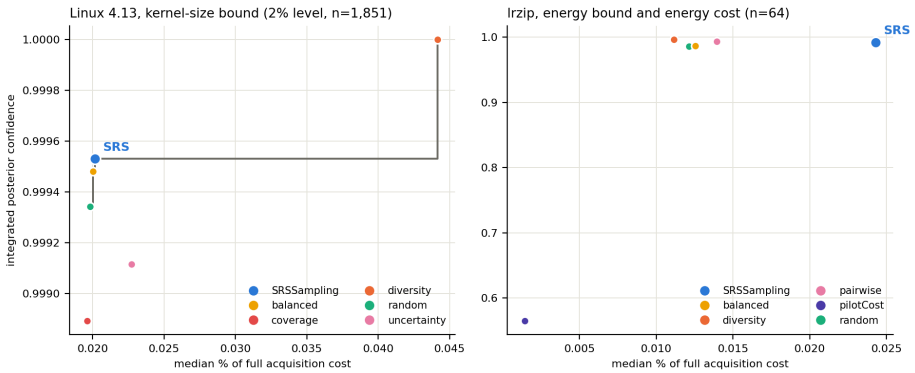


Fig. 6 Guarantee/cost view of two key extension systems at one budget level each (points: strategies; line: empirical cost/confidence Pareto frontier; median integrated confidence versus median share of full acquisition cost). Left: Linux 4.13, kernel-size bound (pool-relative; note the narrow confidence range). Right: lrzip with energy as both target and cost.

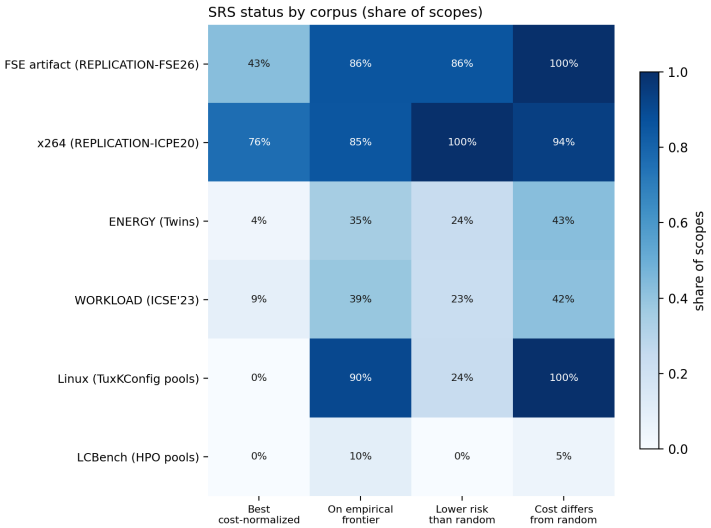


Fig. 7 SRS status per corpus under measured cost (share of scopes).

ysis does not dismiss the strategy, it *prices* it. Figure 6 shows two key extension systems in detail. On the Linux 4.13 pool (92,562 kernels, 2% budget level), SRS is the archetypal frontier point: essentially the best confidence available at its spend, while diversity buys the final $5 \cdot 10^{-4}$ of integrated confidence at more than twice the cost. On the lrzip energy scenario, SRS is dominated—diversity achieves more confidence at half the spend—and the cost-aware pilotCost is by far the cheapest strategy yet delivers drastically weaker guarantees, previewing the equal-spend mechanism of Section 5.4.

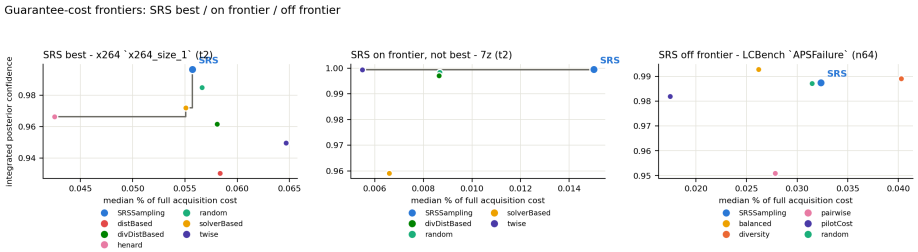


Fig. 8 Representative guarantee/cost frontiers (points: strategies at one budget level; line: empirical cost/confidence Pareto frontier). Left: SRS best (x264). Middle: SRS on the frontier but not best (7z, FSE artifact). Right: SRS dominated (LCBench pool).

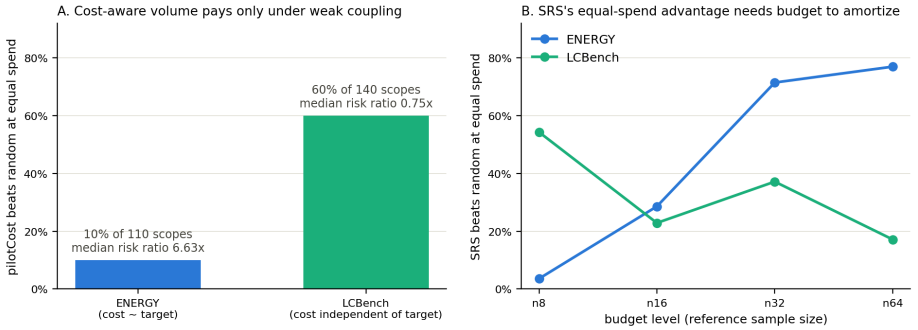


Fig. 9 Equal-measured-spend experiments. (A) pilotCost versus uniform random across the coupling spectrum: share of scopes where pilotCost's realized bound risk is lower, with the median risk ratio. (B) Share of scopes where SRS beats random at equal spend, by budget level.

Finding 7. Under cost normalization SRS keeps its crown only on the replication corpora (x264 26/34; FSE artifact 3/7). On Linux it is never best yet frontier-competitive in 90% of scopes with an always-significant cost premium: cost-indexing prices strategies rather than dismissing them.

5.4 RQ2.5: Equal spend, and the mechanism

The ω_{cost} handicap is a counterfactual. The equal-spend protocol (Section 3.6) makes it literal: every strategy spends the same measured budget B and buys however many configurations B affords. This is exactly the setting that cost-aware strategy design targets (Sarkar et al., 2015; Nair et al., 2017), and pilotCost is built for it: it converts a fixed spend into substantially more measured configurations than any cost-blind strategy.

For guarantees, the picture splits by *cost–target coupling* (Figure 9A). On ENERGY, where cost (energy or runtime) is strongly correlated with the bounded target, pilotCost converts its budget into 2.05× more configurations (median; up to 5× on cost-heterogeneous systems)—and still loses: it beats random in 11/110 scopes, with a median realized risk 6.6× *worse*. The information that a worst-case bound needs lives in the expensive tail of the cost distribution, precisely the region a cost-avoiding strategy refuses to buy; no volume of cheap configurations substitutes for it. On LCBench, where training runtime and final accuracy are only weakly coupled, the verdict flips: with a 1.66× volume advantage, pilotCost beats

Table 8 RQ2.5, equal measured spend: every strategy buys the sample size the budget B affords. Volume: pilotCost's median affordable-n multiplier over random. Risk ratio: median of pilotCost's realized bound risk over random's.

		pilotCost vs. random				
Family	Scopes	volume	beats	risk ratio	SRS beats random	Top winner
ENERGY (cost $\sim$ target)	110	2.05×	11 (10%)	6.6×	49 (45%)	diversity (47)
LCBench (cost $\perp$ target)	140	1.66×	84 (60%)	0.75×	46 (33%)	diversity (80)

random in 84/140 scopes with a median risk ratio of 0.75. Cost-aware volume pays exactly when the outcome-relevant information is *not* concentrated in the expensive tail.

Table 8 summarizes both experiments.

The same experiment resolves the SRS budget question raised in RQ_1 (Figure 9B). On ENERGY, SRS's equal-spend win rate over random climbs from 1/28 scopes at n8-scale budgets to 20/26 at n64-scale: its recursive space restriction needs enough batches to pay for itself, which is why it dominates at the original study's t-wise-scale budgets and disappears at small ones. On LCBench the gradient runs the other way (54% down to 17%): with cost and target decoupled, random's cheap volume catches up with the pool's extremes faster than SRS's guided-but-expensive search. Across both experiments the overall winner is unchanged—diversity wins 47/110 and 80/140 scopes—so the equal-spend protocol and the ω_{cost} handicap agree on every corpus-level conclusion, ruling out the methodology as the explanation.

Finding 8. At literal equal spend, cost-effectiveness is governed by cost–target coupling and budget amortization: the cost-aware pilotCost flips from $6.6\times$-worse to better-than-random as coupling weakens, and SRS's advantage grows from 4% to 77% of scopes as the budget grows. "Cost-effective" is a property of the (strategy, outcome, coupling) triple—not of a strategy alone.

5.5 RQ_2 answer

The FSE metrics themselves survive the re-indexing unchanged: posterior risk, confidence, and ω are as computable and discriminative against a cost-indexed baseline as against a size-indexed one. What does not survive is the invariance of the *decisions* they support. The frontier of defensible choices moves in 97.5% of scopes; the cost-normalized winner is SRS only within the original population; and equal-spend testing shows the governing mechanisms to be properties the sample-size abstraction cannot express—where cost lives relative to the target, and how much budget a strategy needs before its guidance amortizes.

6 Discussion

6.1 What kind of replication outcome is this?

Our results are simultaneously a strong confirmation and a strong qualification of Weis et al. (2026b), and it is worth being precise about which is which. Everything the original paper *measures* replicates: the guarantee framework discriminates strategies wherever we take it, its statistical behavior is robust to FDR correction, and on the original artifact—with the authors' own strategy samples—every headline claim reproduces under an independent

reimplementation. What does not generalize is the *decision layer* built on top: which strategy to recommend, and at what budget. Both turned out to be population-relative in ways the original single-corpus, single-budget-scale design could not observe. This is precisely the epistemic value that replication studies are supposed to add (Shull et al., 2008; Gómez et al., 2014): not a verdict on the original, but a map of where its conclusions hold.

6.2 The triple: strategy, outcome, coupling

The equal-spend experiments turn the cost argument from an accounting observation into a mechanism. In learning-oriented settings, cost-aware sampling is defensible: cheap configurations carry option-effect information a learner can use, so buying more of them per unit spend can be a real gain (Sarkar et al., 2015; Nair et al., 2017). For worst-case bounds the same policy can be self-defeating, because a bound is an extreme order statistic: only samples near the top of the target distribution move it, and when acquisition cost correlates with the target, those are exactly the configurations a cost-avoider skips. The pilotCost flip (risk ratio 6.6× worse under strong coupling, 0.75× under weak) is the cleanest demonstration we know of that *cost-effectiveness is not a strategy property*: the same strategy, under the same equal-spend accounting, is the right choice for one (outcome, coupling) pair and the wrong one for another. Evaluations that report a single "best strategy per dollar" without fixing the outcome and characterizing the coupling are underdetermined.

The SRS budget gradient is the second face of the same coin. SRS spends its early budget learning *where* to sample; that investment amortizes only if enough budget remains to exploit it. At the original study's t-wise-scale budgets it does; at n=8–64 it does not. Neither our small-budget result nor the original large-budget result is wrong—they sit at different points of an amortization curve that Figure 9B makes explicit.

6.3 Implications

For practitioners. First, budget in measured units, not samples: the frontier of defensible choices moves in 97.5% of scopes when the axis switches, and same-size samples can differ in real spend by an order of magnitude. Second, think in frontiers, not winners: the single top recommendation is unstable across corpora, but frontier membership is informative—on Linux, SRS is never the winner yet almost always worth considering, at a quantifiable premium. Third, match the strategy to the deliverable: space-filling strategies (diversity, pairwise) are the robust default for small-budget bounds; SRS earns its cost at large budgets on configurable software; cost-aware selection helps only when the target and the cost are decoupled.

For researchers evaluating sampling strategies. Posterior risk/confidence deserves adoption as a standard outcome: it is learner-free, cheap to compute on measured universes, and—as our 851-scope study shows—transportable. But evaluations should report the measured-cost axis alongside sample size (both are available in every corpus we used at zero additional measurement cost), state the cost–target coupling of their subjects, and span more than one budget scale before issuing strategy recommendations. Our results suggest a reporting template: guarantees per budget under both axes, frontier membership rather than single winners, and equal-spend spot checks for any cost-aware strategy.

For the FSE 2026 framework itself. Nothing in our data challenges the framework; two extensions would widen its reach. First, a cost-indexed prior (N_{eff}, Section 2.3) is a drop-

in addition that turns the existing ω into a cost-normalized decision criterion. Second, the pool-relative variant we used for Linux and LCBench—posterior risk with respect to a large measured pool rather than an exhaustive universe—extends the framework to systems whose spaces cannot be enumerated, at the price of re-scoping the guarantee to the pool.

6.4 Why the guarantee setting is the right test bed

Had we observed these effects on a prediction-error outcome, a skeptic could attribute them to learner idiosyncrasies—regressor choice is known to interact with sampling (Kaltenecker et al., 2020). Posterior risk removes that escape: no learner is involved, so every decision change under cost re-indexing is attributable to the cost construct itself. The guarantee setting is also where the practical stakes are highest—bounds are typically bought under explicit measurement budgets—and where the mechanism is most transparent, because an extreme order statistic makes the location of "useful information" analytically obvious.

7 Threats to Validity

Construct validity. Our cost construct κ is the measured value of a cost-like property shipped with each corpus (runtime, energy, training time). It captures the dominant marginal cost of acquiring one configuration's measurement but omits fixed costs (build and deployment machinery, engineer time) and possible parallelism. On Linux, cost and target coincide (kernel size proxies build effort), which makes the coupling maximal by construction; we therefore base no coupling-mechanism claim on Linux and derive the mechanism from the ENERGY/LCBench contrast, where cost and target are measured independently. Conversely, the equal-spend budgets ($B = n_{\text{ref}} \cdot \bar{\kappa}$) inherit random sampling's expected spend as the reference; other anchors would shift absolute win rates but not the ENERGY-versus-LCBench contrast.

Internal validity. All pipelines are seeded and were verified to be deterministic end to end (an interrupted Linux sweep reproduced its completed part bit-identically when rerun). Significance-based results are FDR-controlled per (corpus, hypothesis) family; the headline H1 counts move only on LCBench ($154 \rightarrow 144$ of 210). Stochastic strategies use 30–100 seeds; deterministic strategies contribute single runs, so their per-scope comparisons rest on decision rules rather than tests. Our SRS reimplementation was audited against the artifact (Section 3.3); the FSE-artifact corpus itself uses the authors' shipped samples, insulating the close replication from reimplementation bias. The artifact's duplicate-containing SRS fills mean its nominal sample sizes are occasionally 1–5 configurations optimistic; this slightly favors SRS in size-indexed comparisons and is therefore conservative with respect to our cost-indexed conclusions.

External validity. Linux and LCBench are large measured pools, not exhaustive configuration spaces; every posterior-risk statement for them is pool-relative, and pool-construction bias (e.g., how TuxKConfig configurations were generated) bounds what the guarantees mean. LCBench is deliberately outside configurable software; we use it to probe the framework's domain sensitivity, not to make claims about hyperparameter optimization practice. Our budget grids emphasize small samples on the generated families ($n \leq 64$, $\leq$30% of pools), the regime where practitioners with expensive measurements operate but also the

regime least favorable to SRS; the x264 and FSE-artifact corpora cover the original t-wise-scale regime, and the equal-spend gradient connects the two. We study worst-case (maximum) bounds only; best-case bounds and the original study's measurement-noise and restricted-ground-truth robustness experiments are not replicated here.

Conclusion validity. Counts aggregate scopes, and scopes within a corpus share systems and pools; we therefore never pool scopes across corpora for significance testing and report per-corpus counts throughout. Where we summarize across corpora (e.g., 830/851), the figure is descriptive, not inferential.

8 Related Work

Sampling and performance learning of configurable systems. A large body of work learns performance models from sampled configurations (Guo et al., 2013; Siegmund et al., 2015; Guo et al., 2018; Ha and Zhang, 2019; Dorn et al., 2023), surveyed by Pereira et al. (2021). Strategy comparisons in this line—distance-based sampling (Kaltenecker et al., 2019), its interplay with learners (Kaltenecker et al., 2020), sampling effects on prediction (Pereira et al., 2020), comparisons of constraint-aware samplers (Medeiros et al., 2016), uniform SAT-sampling (Plazar et al., 2019; Heradio et al., 2022), and coverage oriented generators (Baranov and Legay, 2024; Heß et al., 2024)—almost universally budget by sample size. Cost-awareness appears as a strategy *design* goal (Sarkar et al., 2015; Nair et al., 2017, 2020) rather than as an *evaluation* axis; the present paper makes the evaluation-axis argument, for the outcome—guarantees—where it can be tested without a learner in the loop.

Guarantees for sampled configurations. Scenario theory (Calafiore and Campi, 2006; Campi and Garatti, 2008) underpins probabilistic guarantees for sampled decision problems and has been applied to testing self-adaptive systems (Mandrioli and Maggio, 2020) and to near-optimal configuration search with statistical guarantees (Oh et al., 2017, 2023). Weis et al. (2026b) bring this lens to strategy *comparison* via posterior risk and confidence; ours is, to our knowledge, the first independent replication of that study and the first evaluation of any guarantee framework under measured acquisition cost.

Replication in empirical software engineering. Replications remain rare relative to their epistemic value (Shull et al., 2008; Gómez et al., 2014). Our design follows the recommended pattern of combining a close replication (original artifact, exact data reuse, independent analysis) with conceptual replications that vary populations and one construct at a time, so that agreement and divergence are both interpretable. Transfer studies across environments and workloads (Jamshidi et al., 2017; Mühlbauer et al., 2023) play a complementary role: they vary the measurement context, whereas we vary the budget construct.

9 Conclusion

We replicated the FSE 2026 posterior-guarantee study of configuration sampling strategies across six corpora and 851 decision scopes, and re-indexed its decisions by measured acquisition cost. The framework replicates without qualification: posterior risk and confidence discriminate strategies in $\approx$95% of scopes, on targets from encoding time to kernel size to model accuracy, with no learner in the loop. The recommendations attached to it are

population-relative: SRS's superiority holds at the original study's budget scale on configurable software and vanishes elsewhere; under measured cost it is never the best choice outside the replication corpora, though often a defensible one at a quantifiable premium.

The cost re-indexing yields the paper's central empirical fact—the guarantee/cost Pareto frontier differs from the sample-size frontier in 97.5% of scopes—and the equal-spend experiments explain the recommendation instability mechanistically: cost–target coupling determines whether cheap volume can substitute for expensive information, and budget scale determines whether guided search amortizes. A counted sample is not a faithful budget for guarantees; and "cost-effective" is not a property a sampling strategy has, but one it exhibits relative to an outcome and a coupling. We release all code, result tables, and a command-level manifest to make both the replication and the extension independently checkable.

A Per-Release and Per-System Breakdowns

Tables 9 and 10 unfold two corpora that the main text reports only in aggregate. Both tables are generated directly from the released result CSVs by a script shipped in the replication package.

The Linux breakdown (Table 9) shows that the corpus-level story is not driven by particular releases: strategy distributions differ everywhere, SRS sits on the empirical frontier in 19 of 21 scopes without ever winning the cost-normalized comparison, and the frontier itself changes at every scope (all Jaccard overlaps ≤ 0.20). diversity wins 19 of 21 scopes; both exceptions (uncertainty on release 4.13, balanced on 4.15) occur at the 5% level, the largest sample regime.

The ENERGY rollup (Table 10) shows the same homogeneity across systems: no system is "safe" for the sample-size abstraction—the frontier changes in every scope of all 14 systems—and the most frequent cost-normalized winner is a space-filling strategy (diversity or pairwise) on all of them. Strategy distributions differ in every scope except on brotli (4 of 12), the smallest table (180 configurations after revision pinning). SRS is best cost-normalized only on LLVM, MongoDB, and jump3r scopes, all at the two largest budgets, consistent with the amortization gradient of Section 5.4.

Acknowledgements

Declarations

Funding. Not applicable.

Conflict of interest. The authors declare that they have no conflict of interest.

Ethics approval and consent to participate. Not applicable.

Data availability. All datasets used in this study are publicly archived: the original FSE 2026 artifact (Weis et al., 2026a), the distance-based sampling and x264 corpora (Kaltenecker et al., 2019; Pereira et al., 2020), the energy corpus (Weber et al., 2023), the workload corpus (Mühlbauer et al., 2023), the TuxKConfig pools (Martin et al., 2022), and LCBench (Zimmer et al., 2021).

Code availability. All experiment code, per-corpus result tables, hypothesis tests, figures, and a command-level run manifest (including data provenance and checksums) are available in our replication package.

Author contributions.

Table 9 Per-release breakdown of the full Linux protocol (pool-relative; levels 1%/2%/5% of each release's measured pool). *n*: sample size at the level; H1: posterior-risk distributions differ (Kruskal–Wallis, α=.05); winner: best cost-normalized strategy (ω_{cost}); SRS front.: SRS on the empirical cost/confidence Pareto frontier; top chg.: top recommendation changes under the cost axis; *J*: Jaccard overlap of the two frontiers.

Release	Level	*n*	Winner	H1	SRS front.	Top chg.	*J*
413	1%	926	diversity	✓	✓	✓	0.14
413	2%	1,851	diversity	✓	✓	–	0.14
413	5%	4,628	uncertainty	✓	✓	✓	0.17
415	1%	394	diversity	✓	✓	✓	0.14
415	2%	788	diversity	✓	✓	✓	0.17
415	5%	1,970	balanced	✓	–	✓	0.20
420	1%	235	diversity	✓	✓	✓	0.12
420	2%	470	diversity	✓	✓	–	0.17
420	5%	1,174	diversity	✓	✓	–	0.17
500	1%	200	diversity	✓	✓	✓	0.14
500	2%	399	diversity	✓	✓	–	0.14
500	5%	998	diversity	✓	✓	–	0.12
504	1%	258	diversity	✓	✓	✓	0.17
504	2%	517	diversity	✓	✓	–	0.12
504	5%	1,292	diversity	✓	✓	–	0.14
507	1%	202	diversity	✓	✓	✓	0.14
507	2%	403	diversity	✓	–	✓	0.17
507	5%	1,008	diversity	✓	✓	✓	0.14
508	1%	219	diversity	✓	✓	✓	0.12
508	2%	438	diversity	✓	✓	✓	0.14
508	5%	1,096	diversity	✓	✓	–	0.17

Table 10 Per-system rollup of the ENERGY family (each system contributes up to 16 scopes: {energy, runtime} cost × {energy, runtime} target × budgets *n*8–*n*64, protocol-capped). H1 / SRS best / SRS front. / top chg. / frontier chg.: scopes where the outcome holds; winner: most frequent best cost-normalized strategy (scopes won).

System	$\lvert\mathscr{V}\rvert$	Scopes	H1	SRS best	SRS front.	Top chg.	Front. chg.	Winner
7z	68,640	16	16	0	4	1	16	pairwise (9)
Apache	13,440	16	16	0	0	0	16	diversity (12)
brotli	180	12	4	0	2	4	12	diversity (8)
exastencils	86,058	16	16	0	8	0	16	pairwise (10)
HSQLDB	864	16	16	0	2	0	16	diversity (14)
jump3r	4,296	16	16	1	10	12	16	pairwise (5)
kanzi	3,202	16	16	0	10	4	16	pairwise (12)
LLVM	65,536	16	16	4	6	0	16	pairwise (10)
lrzip	5,184	16	16	0	7	2	16	diversity (14)
MongoDB	6,840	16	16	4	6	0	16	pairwise (6)
nginx	4,416	16	16	0	7	6	16	diversity (8)
PostgreSQL	864	16	16	0	5	0	16	pairwise (8)
VP8	41,040	16	16	0	6	0	16	diversity (16)
x264	4,608	16	16	0	4	2	16	diversity (12)

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